Walker AI - Material Cost Library (MVP) - Version 3

This version expands the material layer to support both pricing and procurement accuracy. It keeps materials separate from labor, adds explicit consumables, includes ordering-focused items such as tile spacers and caulking/sealants, and introduces a waste-factor model so the same engine can later drive supplier integrations, contractor shopping lists, and DIY kits.

Core principle

Walker AI should track every relevant material item for ordering, even if some items have small cost impact. The system should therefore maintain two linked outputs from the same quantity engine: (1) a pricing output for estimate calculation, and (2) a detailed material list for procurement and ordering.

Material Cost Library

Category	Material Item	Unit	Low	Average	High
Tile / Surface	Floor tile material	SQF	$4	$7	$12
Tile / Surface	Wall tile material	SQF	$5	$8	$15
Tile / Surface	Shower tile material	SQF	$6	$10	$18
Tile / Surface	Backsplash tile material	SQF	$5	$9	$16
Tile / Surface	Grout	Bag Eq.	$18	$28	$45
Tile / Surface	Thinset / mortar	Bag Eq.	$22	$35	$55
Tile / Surface	Tile edging / trim	LF	$3	$5	$8
Tile / Surface	Tile leveling clips / system	Box	$12	$22	$38
Tile / Surface	Tile spacers	Box	$6	$10	$18
Tile / Surface	Silicone sealant (wet area)	Tube	$10	$16	$28
Tile / Surface	Interior caulking (paintable)	Tube	$8	$12	$20
Waterproofing	Waterproof membrane	SQF	$2	$3.5	$5.5
Waterproofing	Liquid waterproofing	SQF	$1.5	$2.5	$4.0
Waterproofing	Shower pan kit	EA	$180	$280	$450
Waterproofing	Shower niche waterproof kit	EA	$45	$75	$140
Waterproofing	Waterproof tape / corners	EA	$20	$35	$60
Drywall	Drywall board	Sheet	$16	$22	$35
Drywall	Moisture resistant drywall board	Sheet	$20	$28	$40
Drywall	Cement board	Sheet	$18	$25	$38
Drywall	Joint compound	Bag	$18	$25	$40
Drywall	Drywall screws / fasteners	Box	$15	$25	$40
Insulation	Batt insulation	SQF	$1.2	$2.0	$3.4
Insulation	Rigid foam insulation	SQF	$1.8	$2.8	$4.5
Insulation	Acoustic insulation	SQF	$1.5	$2.4	$3.8
Framing	2x4 framing lumber	LF	$1.8	$2.8	$4.5
Framing	2x6 framing lumber	LF	$2.6	$3.8	$6.0
Framing	Pressure treated lumber	LF	$3.2	$4.8	$7.5
Framing	Plywood / sheathing	Sheet	$28	$42	$65
Framing	OSB	Sheet	$22	$34	$52
Framing	LVL / beam material allowance	LF	$18	$28	$48
Framing	Construction adhesive / connectors	EA	$18	$30	$55
Framing	Wood shims	Bundle	$4	$8	$16

Category	Material Item	Unit	Low	Average	High
Flooring	Laminate flooring material	SQF	$2.0	$3.5	$6.0
Flooring	Vinyl plank flooring material	SQF	$2.5	$4.25	$7.0
Flooring	Engineered hardwood material	SQF	$4.5	$7.5	$12.0
Flooring	Tile underlayment	SQF	$1.0	$1.8	$3.0
Flooring	Flooring underpad	SQF	$0.5	$0.9	$1.6
Flooring	Self-leveling compound	Bag Eq.	$28	$42	$68
Flooring	Heated floor mat / cable	SQF	$7.0	$11.0	$18.0
Paint	Interior wall paint	Gallon Eq.	$25	$40	$70
Paint	Ceiling paint	Gallon Eq.	$22	$36	$60
Paint	Primer	Gallon Eq.	$18	$28	$45
Paint	Trim / cabinet paint materials	EA	$20	$35	$60
Paint	Caulking / sealant	Tube	$8	$12	$20
Paint	Sandpaper / prep materials	EA	$8	$15	$28
Paint	Wood filler / patching compounds	EA	$8	$16	$30
Plumbing Fixtures	Standard toilet allowance	EA	$250	$425	$700
Plumbing Fixtures	Vanity allowance	EA	$400	$850	$1500
Plumbing Fixtures	Sink / basin allowance	EA	$120	$240	$500
Plumbing Fixtures	Sink faucet	EA	$90	$180	$350
Plumbing Fixtures	Shower valve / trim allowance	EA	$150	$300	$650
Plumbing Fixtures	Shower head / trim kit	EA	$80	$180	$300
Plumbing Fixtures	Tub filler / faucet set	EA	$140	$320	$700
Plumbing Fixtures	Bathtub allowance	EA	$500	$1100	$2200
Plumbing Fixtures	Shower base / pan allowance	EA	$180	$350	$750
Plumbing Materials	PEX / supply material	LF	$2.5	$4.0	$6.5
Plumbing Materials	ABS / PVC drain material	LF	$4.0	$6.5	$10.5
Plumbing Materials	Shutoff valves / angle stops	EA	$20	$40	$80
Plumbing Materials	Flexible supply lines	EA	$10	$18	$35
Plumbing Materials	Trap / drain assembly	EA	$25	$55	$120
Electrical Fixtures	Standard light fixture allowance	EA	$50	$140	$350
Electrical Fixtures	Pot light allowance	EA	$18	$35	$75
Electrical Fixtures	Bathroom fan allowance	EA	$90	$180	$380
Electrical Fixtures	GFI receptacle material	EA	$18	$32	$55
Electrical Fixtures	Standard receptacle material	EA	$5	$10	$20
Electrical Fixtures	Switch material	EA	$5	$10	$18
Electrical Fixtures	Smart switch material	EA	$25	$55	$120
Electrical Materials	Electrical wire / cable	LF	$0.9	$1.4	$2.4
Electrical Materials	Breakers / panel materials	EA	$45	$90	$180
Electrical Materials	Boxes / covers / connectors	EA	$6	$12	$25
Electrical Materials	Cover plates	EA	$1.5	$3.5	$7.0
Electrical Materials	Wire connectors / marrettes	EA	$6	$12	$22
Electrical Materials	Electrical staples / support hardware	EA	$4	$8	$15
Cabinetry	Stock kitchen cabinets material	LF	$120	$190	$300
Cabinetry	Semi-custom cabinets material	LF	$180	$280	$420
Cabinetry	Custom cabinets material	LF	$260	$380	$600
Cabinetry	Bathroom vanity cabinet material	EA	$220	$480	$950
Cabinetry	Cabinet hardware	EA	$4	$9	$20

Category	Material Item	Unit	Low	Average	High
Cabinetry	Shelving material	LF	$8	$16	$32
Countertops	Laminate countertop material	LF	$25	$40	$65
Countertops	Quartz countertop material	LF	$60	$95	$150
Countertops	Stone / premium countertop material	LF	$80	$130	$220
Countertops	Backsplash slab / panel material	LF	$20	$40	$85
Doors / Trim	Interior door slab / prehung allowance	EA	$120	$240	$500
Doors / Trim	Baseboard material	LF	$1.2	$2.2	$4.2
Doors / Trim	Casing material	LF	$1.3	$2.4	$4.5
Doors / Trim	Crown / trim material	LF	$1.8	$3.2	$6.0
Doors / Trim	Door hardware	EA	$22	$45	$95
Building Materials	Concrete mix / patch material	Bag	$8	$14	$24
Building Materials	Fasteners / screws / anchors	Box	$12	$22	$40
Building Materials	General adhesive / foam / tape allowance	EA	$18	$30	$55
Building Materials	Protection materials (poly, floor protection)	EA	$25	$50	$100
Building Materials	Disposal / waste materials allowance	EA	$40	$80	$180
Waste / Disposal	Disposal bin rental	EA	$200	$350	$500

Material consumables and ordering logic

Small-cost items should still be tracked because they matter for procurement accuracy. These items may contribute little to total estimate value, but they are necessary for ordering, staging, and site execution.

Item	Recommended Trigger / Rule	Ordering Logic
Tile spacers	If any tile area > 0	1 box up to 100 SQF; 2 boxes for 100-250 SQF; 3 boxes for 250+ SQF
Tile leveling clips	If large-format tile or tile area > 80 SQF	1 box default; increase for larger areas
Silicone sealant	If wet area, shower, tub, sink, backsplash	1-3 tubes depending on fixture count and wet-area size
Interior caulking	If trim, casing, baseboard, paint finish work	1-3 tubes depending on LF of finish trim
Drywall screws	If drywall board quantity > 0	1 box for small rooms; more for larger board counts
Fasteners / anchors	If framing, fixtures, or accessory installs	1 box default per scope unless structural scope is large
Shims	If cabinetry, doors, trim, framing	1 bundle default
Wire connectors	If electrical scope exists	1 pack default; increase with device count
Disposal bin rental	If demolition scope exceeds small repair threshold	1 bin for standard bathroom/kitchen demo; more for larger scopes

Material waste factor model

To improve ordering accuracy, the quantity engine should apply material-specific waste factors before generating the final material list. Waste should be applied to ordering quantities, and optionally reflected in material pricing where appropriate.

Material Category	Recommended Waste Factor	Notes
Tile	10% standard / 15% complex layouts	Increase for diagonal patterns, niches, many cuts, or small-format mix
Flooring planks	7% standard / 10% complex	Increase for angled rooms and pattern matching
Drywall / cement board	10%	Allows for cutoffs and damaged sheets
Framing lumber	8-10%	Allows for culls, cutoffs, and blocking
Countertops	Handled by slab/fabrication logic	Do not use simple waste percentage only
Paint	10%	Allows for extra coats, porosity, and touch-ups
Waterproof membrane	8-10%	Allows for overlaps, corners, niches, and errors

Material shopping list output model

The estimate engine should generate an internal structured shopping list from the same measured quantities, waste factors, and material library used in pricing. The estimate and shopping list must come from the same data source to avoid mismatches.

Output Field	Example
category	Tile / Surface
item_name	Floor tile material
base_quantity	60 SQF
waste_factor	10%
ordered_quantity	66 SQF
unit_cost_average	$7
line_total	$462

Example shopping list output

- Floor tile: base 60 SQF -> ordered 66 SQF after 10% waste

- Thinset / mortar: 2 bags equivalent
- Grout: 1 bag equivalent
- Tile spacers: 1 box
- Tile leveling clips: 1 box
- Vanity: 1
- Sink faucet: 1
- Shower valve: 1
- Shower head: 1
- Drywall boards: 6 sheets
- Drywall screws: 1 box
- Silicone sealant: 2 tubes
- Disposal bin rental: 1

Recommended implementation rules

1. Keep materials separate from labor in the estimate engine.
2. Use the Average column as default MVP pricing unless a tiered material selection changes it.
3. Track small consumables even when their cost impact is low, because they improve ordering accuracy.
4. Apply waste factors before finalizing ordered quantities.
5. Use the same quantity engine for pricing and shopping-list output.
6. Store material data in a modular material_pricing table and ordering data in a material_list output object.
7. Do not hard-code shopping-list quantities into the UI; derive them from room measurements, scope, and triggers.